WEEKLY ACTIVITY

1) Create a GitHub repository for the application

Ans : create the github repo

And clone in the ubuntu terminal

```
vrushali@Vrushali:~$ git clone https://github.com/Vrushk13/devops-jenkins-activity.git
Cloning into 'devops-jenkins-activity'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

Create shell based application

App.sh

Run this application using the command : ./app.sh

```
vrushali@Vrushali:~/devops-jenkins-activity$ ./app.sh
Hello from Jenkins CI Pipeline
 Jenkins Application build successful
vrushali@Vrushali:~/devops-jenkins-activity$
```

Create the Test file for the application

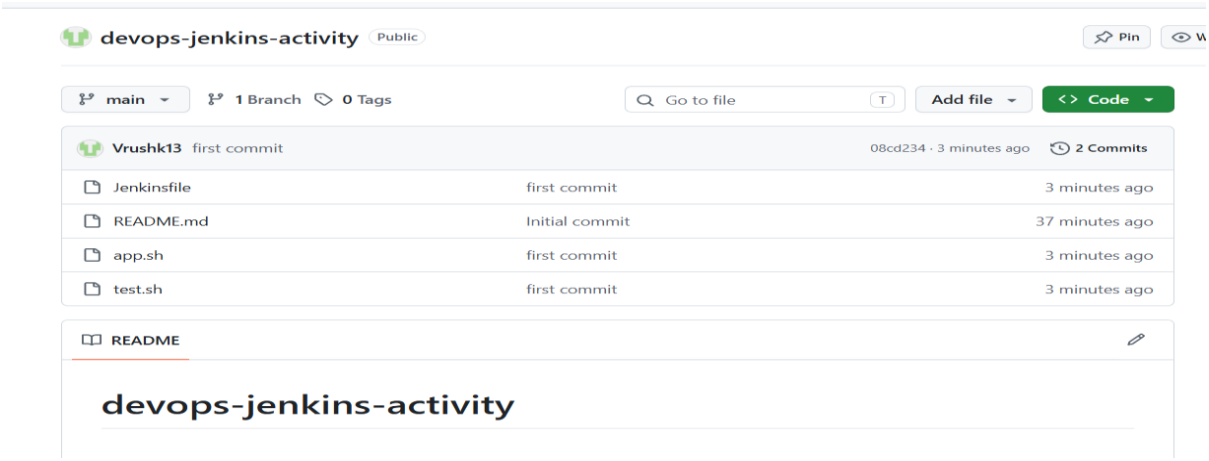
Nano test .sh

```
vrushali@Vrushali:~/devops-jenkins-activity$ nano test.sh
vrushali@Vrushali:~/devops-jenkins-activity$ ls
README.md  app.sh  test.sh
vrushali@Vrushali:~/devops-jenkins-activity$ chmod +x test.sh
vrushali@Vrushali:~/devops-jenkins-activity$ ./test.sh
Running application test...
TEST PASSED: app.sh exists.
vrushali@Vrushali:~/devops-jenkins-activity$
```

Create the Jenkins file to automate the process

Write the script

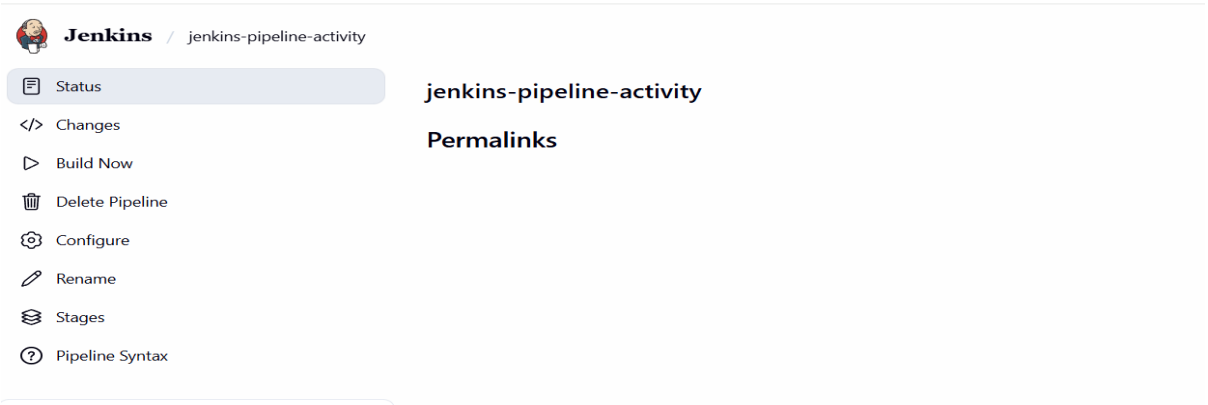
Push the code to the github



2) Connect github to Jenkins

Ans : Connect the github to Jenkins

Configure the github with Jenkins and create the pipeline



After the build the pipeline

```
 > git config core.sparsecheckout # timeout=10
 > git checkout -f 08cd23415da4b702c3ea26548e8f83b4e83fe06d # timeout=10
Commit message: "first commit"
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building application...
[Pipeline] sh
+ chmod +x app.sh
[Pipeline] sh
+ ./app.sh
Hello from Jenkins CI Pipeline
 Jenkins Application build successful
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests...
[Pipeline] sh
+ chmod +x test.sh
```

```
+ ./test.sh
Running application test...
TEST PASSED: app.sh exists.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Validation)
[Pipeline] echo
Validating application...
[Pipeline] sh
+ test -f app.sh
[Pipeline] sh
+ test -f test.sh
[Pipeline] echo
Validation successful!
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI Pipeline completed successfully!
```

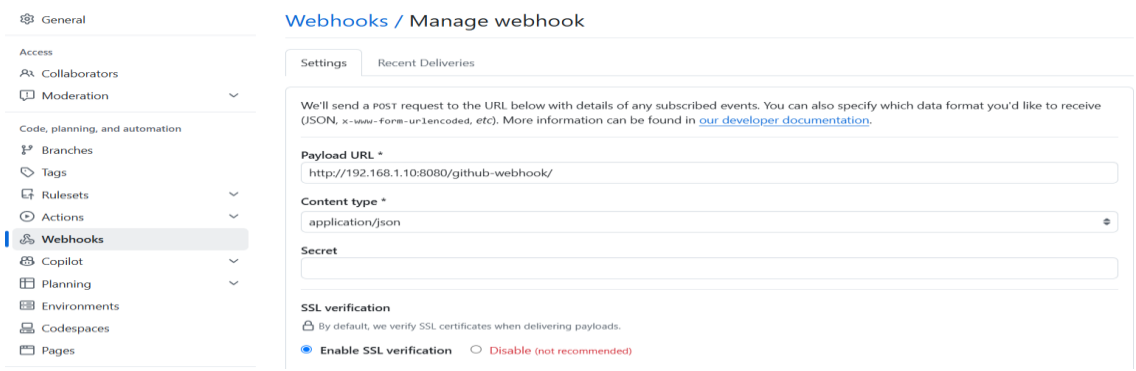
```
[Pipeline] echo
CI Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

3) Configure Github Webhook

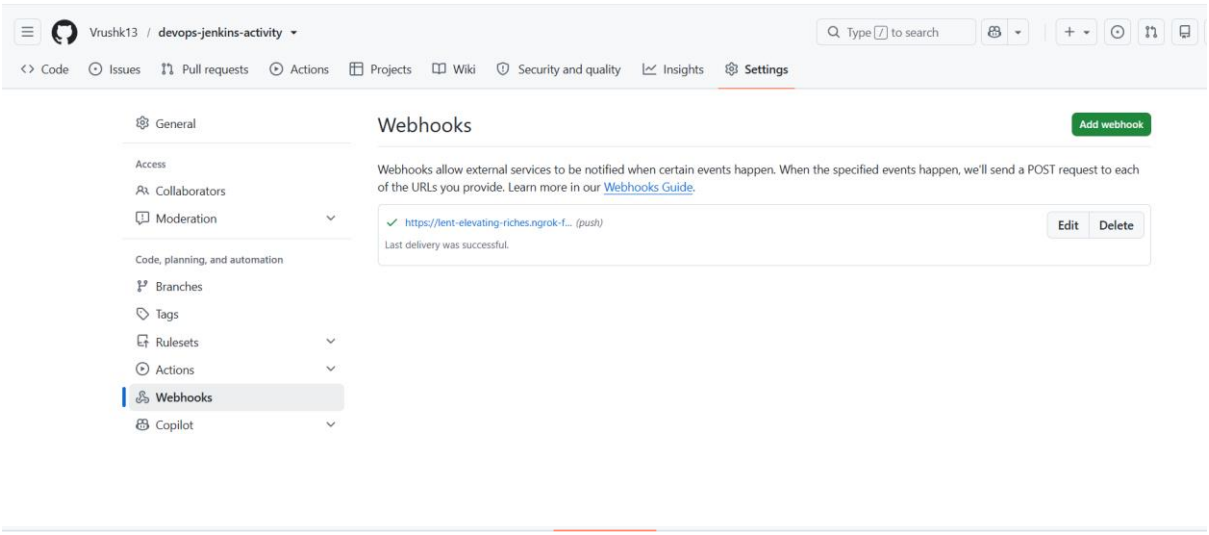
To configure github webhook first we have to build the pipeline successfully

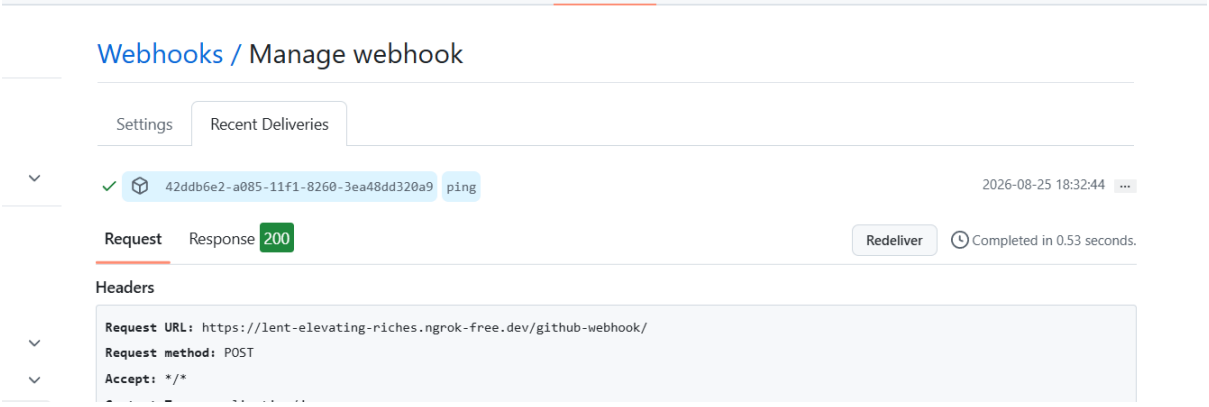
And then configure the github hook

1) In Jenkins configure the github hook

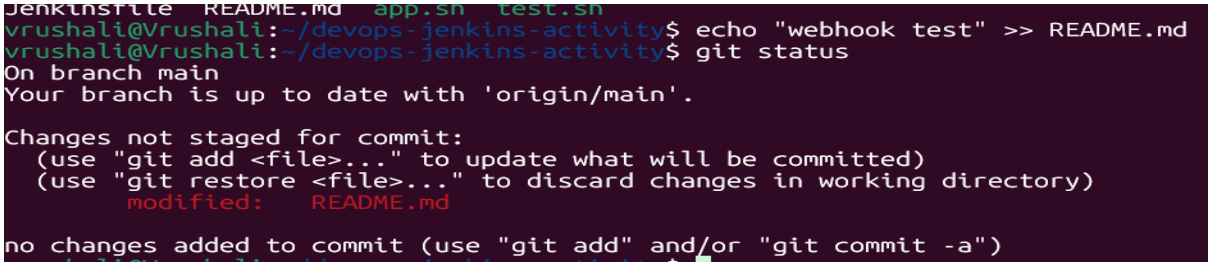


Install the ngrok for Jenkins reach to the pribate IP
Configure it with github



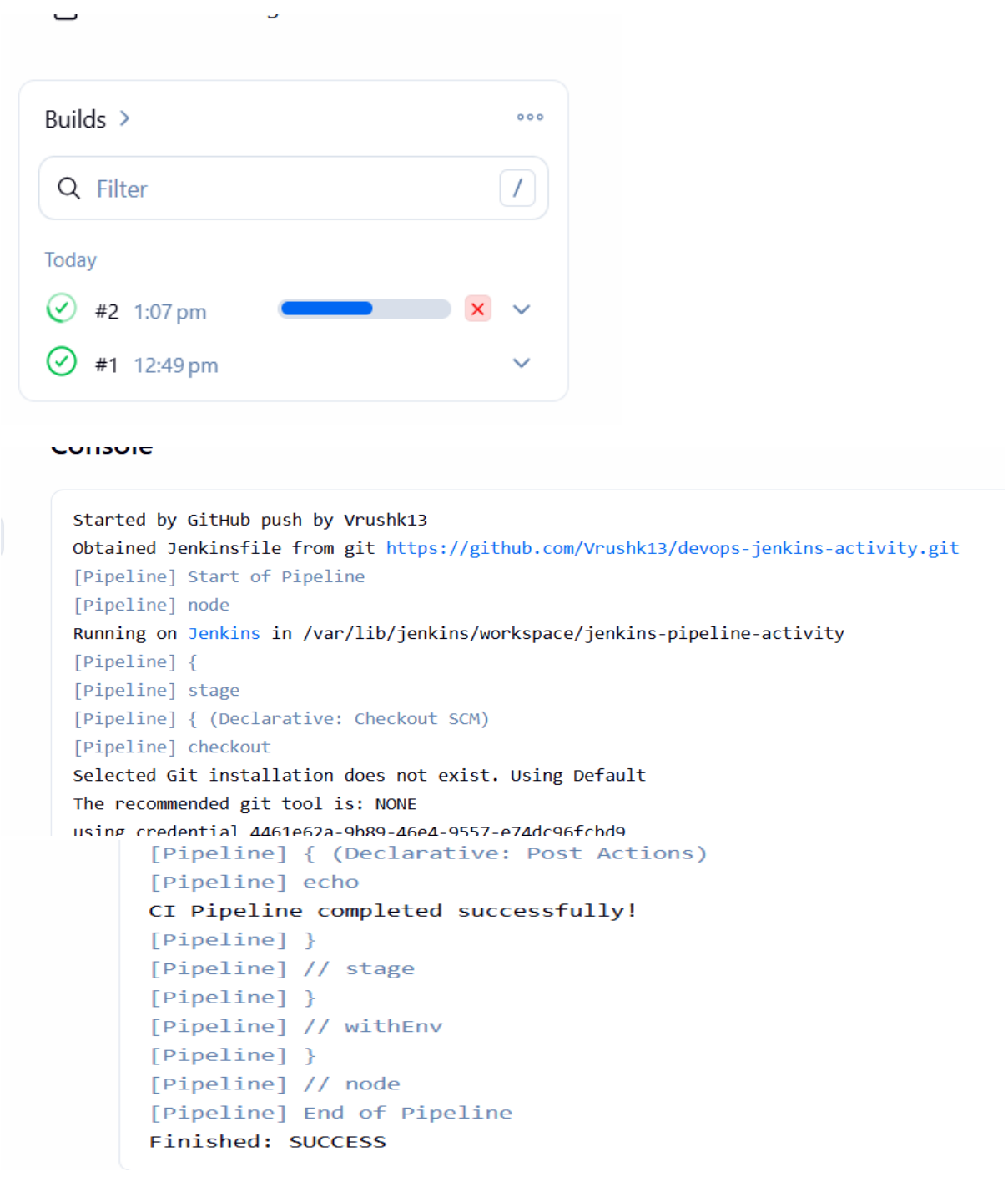


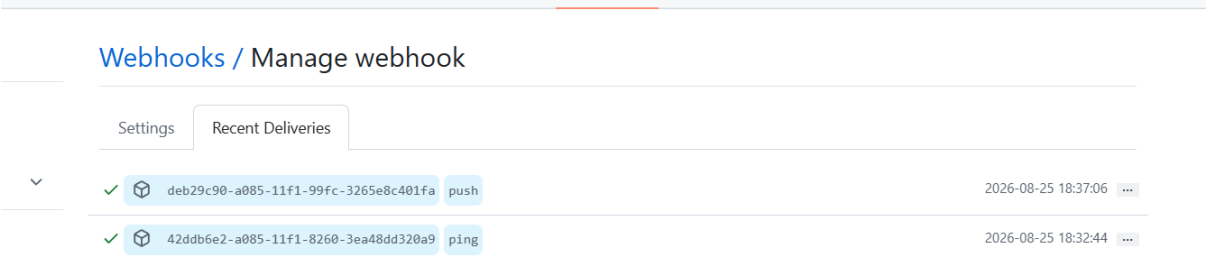
After changes in the code



After changes in the code and push into the github

Pipeline is automatically trigger





4) **Introduce a controlled error into the project.**

Ans: In the test.sh file add the line to fail the pipeline

if [-f wrong-file.sh]; then

And we got the output like

```
+ ./test.sh
Running application test...
TEST FAILED: app.sh does not exist.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Validation)
Stage "Validation" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI Pipeline failed!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
ERROR: script returned exit code 1
Finished: FAILURE
```

5) **Fix the error and run the pipeline again**.

Ans : First , Identify the issue and resolve the issue how to fix the error

I fix the error and run the pipeline

Go to your terminal:

cd ~/devops-jenkins-activity

Open test.sh:

nano test.sh

Find:

if [-f wrong-file.sh]; then

Change it to:

if [-f app.sh]; then

Failure

Test stage failed.

TEST FAILED: app.sh does not exist.

ERROR: script returned exit code 1

Finished: FAILURE

Root cause

The test script was incorrectly checking for

wrong-file.sh instead of app.sh.

Troubleshooting

1. Opened Jenkins failed build.

2. Opened Console Output.

3. Identified the failed Test stage.

4. Found exit code 1.

5. Checked test.sh.

6. Found incorrect file reference.

Resolution

Changed:

wrong-file.sh

to:

app.sh

Short troubleshooting summary

The Jenkins pipeline failed during the **Test** stage because test.sh was checking for the wrong file (wrong-file.sh). I checked the Jenkins Console Output, identified the error and exit code 1, corrected it to app.sh, and pushed the fix to GitHub. Jenkins then automatically triggered the pipeline again, and all stages completed successfully.